The Relationship of Digital Behavior and Mental Health

Data Analysis Report

STAT 641: Applied Linear Models

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Protocol

Collecting data on social behavior and mental wellbeing can be challenging due to privacy concerns related to the release of sensitive health and psychological information. Dataset related to mental health often do not include metrics for digital behavior. In research, mental health data is generally collected through surveys or collected in clinical settings and incorporates various types of psychological illnesses including anxiety, depression, and other personal health information. Such data generally requires IRB approval and other ethics review, which can be time consuming.

For this analysis, the data was retrieved using Kaggle, an online repository platform. Utilizing publicly accessible simulated dataset is cost effective, protects medical and other personal information, and explores a scenario that is not typically collected. Additionally, the dataset bypasses many of the ethical concerns and other restrictions as it is a simulated dataset for public use.

The dataset is titled *Mental Health and Digital Behavior (2020–2024)*. In the dataset, there are nine variables, where five variables measure digital behavior and three mental health indicator variables. Due to the data being simulated and available to the public, IRB approval is not required, no personally identifiable health information is included, and the dataset is free for academic use. The file was downloaded as a csv file and uploaded to SAS for further analysis.

I. Introduction

Historically, social media served as a communication method to connect friends, family, and other individuals. Since the revolution of social media in the early 2000s, there have been numerous platforms created expanding the use of social media (Asher, 2024). Although social media has many beneficial components such as helping build community, developing skills, and interacting with others globally, there are also disadvantages associated with increased usage, bullying, and the dissemination of false information that impact mental illnesses (National Academies of Sciences, Engineering, and Medicine, 2024; González-Padilla & Tortolero-Blanco, 2020).

The purpose of this study is to evaluate the impact of social media usage on mental wellbeing. This study investigates whether increased social media usage is associated with changes in mental wellbeing. Social media has become a daily ritual with a reported 2.1 billion people using social media regularly (Latha et. al., 2020). In this report, we will investigate the digital behaviors that impact wellbeing.

II. Data Source

Data was retrieved from Kaggle which is a public repository. The dataset is artificially generated to replicate digital behavioral patterns generally observed in individuals. The dataset includes digital behavior patterns and mental health indicator variables from 2020-2024. In this study, there were 500 simulated data points and a total of nine variables. Due to the data being artificially generated, it does not represent or include particular individuals or demographics such as race, age, or gender.

Table 1. List of Variables

anxiety_level	mood_score	focus_score
daily_screen_time_min	notification_count	social_media_time_min
digital_wellbeing_score	num_app_switches	sleep_hours

III. Method of Analysis

To examine the relationship between digital behavior and mental wellbeing, a multiple linear regression (MLR) model was used. MLR is the appropriate analysis because the outcome variable is continuous. The goal of the model is to explore the factors and relationship between digital behavior and mental wellbeing. The MLR model is represented below:

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \epsilon$$

Where:

Y_i = outcome/dependent variable

β_0 = intercept

$\beta_1, .., \beta_4$ = regression coefficient

$X_1, ..., X_4$ = predictor/independent variables

ϵ = random error term

For this model, quantitative variables used included the total daily screen time (X_1), time spent using social media (X_2), number of notifications received (X_3), and number of times users switched apps (X_4) as the predictor variables to test the impact on mental wellbeing.The digital wellbeing score is used as the outcome variable in the model. The digital wellbeing score is a calculation that groups the focus, sleep, and anxiety levels, based on self-reported indicators.

The dataset was analyzed with Statistical Analysis System (SAS) and variables of interest were used to compute the descriptive statistics (mean, median, mode, standard deviation, etc.) to provide a summary of the data before performing the regression. Based on the data, we hypothesized that at least one of the variables is significant in predicting digital and mental wellbeing.

Null Hypothesis $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$

Alternative Hypothesis: H_a : At least one $\beta_a \neq 0$

To properly assess whether the regression model was appropriate, we conducted a diagnostics to check the validity and assumptions of the model. The data was then plotted to check the assumptions of the regression model. The Q-Q plot provides evidence that the assumption of linearity, normality, and homoscedasticity were met as the residual plot follows a straight line and there are no cone patterns seen in ***Appendix A.1 - A.4***. The assumption of multicollinearity was met as each predictor variable has a Variance Inflation Factors (VIF) less than 5, no indication of multicollinearity seen in ***Appendix B.2***.

IV. Results

There were a total of 500 observations in the study. Of the variables included in the model, this section reviews the descriptive statistics, model outcomes, and diagnostic determinations concluded by performing the MLR analysis.

On average, participants spent a total of 360.44 minutes with their screen time and 121.77 minutes on social media platforms which can be seen in ***Figure 1***.

Figure 1: Summary Statistics

Variable	Mean	Median	Mode	Minimum	Maximum	Std Dev	Range	Lower Quartile	Upper Quartile	Coeff of Variation	Skewness	Kurtosis
digital_wellbeing_score	52.3104000	51.0000000	48.3000000	34.6000000	80.8000000	7.9431858	46.2000000	46.8000000	56.6000000	15.1847162	0.8002025	0.5398704
daily_screen_time_min	360.4378000	360.8000000	324.0000000	180.0000000	591.2000000	58.7824755	411.2000000	317.9500000	398.3500000	16.3086323	0.1946938	0.2264368
social_media_time_min	121.7718000	119.5500000	30.0000000	30.0000000	296.7000000	43.8094428	266.7000000	91.8500000	149.7000000	35.9766734	0.2683036	0.2183807
notification_count	79.5120000	80.0000000	75.0000000	20.0000000	158.0000000	23.9797396	138.0000000	63.0000000	96.0000000	30.1586422	-0.0103510	-0.0403525
num_app_switches	49.8400000	50.0000000	53.0000000	30.0000000	72.0000000	7.3409931	42.0000000	44.0000000	55.0000000	14.7291195	0.2110271	-0.0266950

By performing a regression analysis, we found that each predictor variable has a p-value < 0.05, which indicates that the predictor variables are statistically significant in predicting the digital wellbeing. Although all variables were identified as being statistically significant, the time spent on social media and notification count variables had a high variance indicating less reliability as predictor variables. Based on our model, a high number of daily screen time, social media time, notifications, and interactions/switching of apps negatively impact the digital wellbeing (i.e., focus, sleep, and anxiety).

Research has shown that prolonged interactions with applications such as Facebook, Twitter, Instagram, YouTube, online gaming, and virtual reality had negative effects on self-esteem, increased severity in anxiety and depressive episodes, and feelings of social isolation (Rideout & Fox, 2018; Hoare et. al., 2016). This analysis continues to support the negative relationship of high social media usage on mental wellbeing. This is evident in the calculation of the confidence interval (CI) as the CI is negative for all predictor variables in ***Table 2***.

Table 2: Model Parameter Estimate with Confidence Intervals

Parameter Estimates								
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation	95% Confidence Limits	
Intercept	1	78.15292	3.19124	24.49	<.0001	0	71.88287	84.42297
daily_screen_time_min	1	-0.01441	0.00532	-2.71	0.0070	1.00297	-0.02486	-0.00396
social_media_time_min	1	-0.04909	0.00713	-6.89	<.0001	1.00045	-0.06309	-0.03509
notification_count	1	-0.12953	0.01304	-9.94	<.0001	1.00286	-0.15514	-0.10391
num_app_switches	1	-0.08774	0.04254	-2.06	0.0397	1.00065	-0.17131	-0.00416

To further investigate the validity of the model, a Cook's Distance (D), DFBETAS, R-Student, Difference in Fits (DDFITS), and Residual test were used to identify the outliers and influential observations. Cook's D identified 29 influential points, DDFITS identified 22 influential observations (> 0.2), and R-Student identified six outliers. To reduce the influence of outliers and influential observations the Robustreg procedure was applied. The model parameter estimates were adjusted to reveal the true trend by downweighting the variables. The predicator variable for the number of times users switched apps is not statistically significant. The procedure also identified seven observations that were unusual and adjusted to not impact the model.

V. Conclusion

Research related to the topic is available and concludes with similar results to the results of the model that increased social media usage can negatively impact the mental wellbeing of an individual. Despite the data being simulated, it does imitate real-world patterns. This analysis revealed that total daily screen time, time spent on social media platforms, and notification count are significant. Recommendations for further research would include additional variables such as age, race, gender, self reporting of anxiety, depression, and other social indicators. It is recommended that reducing the time spent on social media can be beneficial to the overall mental health and wellbeing by incorporating lifestyle changes (i.e., exercising, reading a book).

References

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Appendix

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/*
daily_screen_time_min = Total daily screen time
num_app_switches = Number of times the user switched apps in a
day
sleep_hours    = Number of hours slept per day
notification_count = Number of notifications received
social_media_time_min = Time spent on social media platforms
focus_score = Self-reported focus score (1-10)
mood_score = Self-reported mood score (1-10)
anxiety_level =     Self-reported anxiety level (1-10)
digital_wellbeing_score  = Calculated score based on focus,
sleep, and anxiety levels
*/
OPTIONS VALIDVARNAME = ANY;
PROC IMPORT OUT=health

DATAFILE="/home/u63946554/mental_health_digital_behavior_data.cs
v"
   DBMS=CSV
   REPLACE;
   GETNAMES=YES;
RUN;

/*Reviewing the contents of the dataset*/
proc contents data=health;
run;
/*Descriptive statistics of dataset
          Mean, median, standard deviation, and mode*/
/*proc univariate data = health;
     var digital_wellbeing_score daily_screen_time_min
social_media_time_min notification_count num_app_switches;
run;*/

proc means data = health  mean median mode min max stddev range
q1 q3 cv skewness kurtosis;
     var digital_wellbeing_score daily_screen_time_min
social_media_time_min notification_count num_app_switches;
run;
```

```
/*Scatter plot for the variables*/
proc sgplot data= health;
     scatter x=daily_screen_time_min y=digital_wellbeing_score;
     title "Appendix A.1";
run;

proc sgplot data= health;
     scatter x=social_media_time_min y=digital_wellbeing_score;
          title "Appendix A.2";
run;

proc sgplot data= health;
     scatter x=notification_count y=digital_wellbeing_score;
          title "Appendix A.3";
run;

proc sgplot data= health;
     scatter x=num_app_switches y=digital_wellbeing_score;
          title "Appendix A.4";
run;

/*****Part B: Perform Statistical Analysis******/
/*Testing the significance of the model*/
                              /**which specific predictors
matter, and how...**/
proc reg data=health;
     model digital_wellbeing_score = daily_screen_time_min
social_media_time_min notification_count num_app_switches;
          title "Appendix B.1";
run;

/*Calculation of confidence intervals*/
proc reg data=health;
     model digital_wellbeing_score = daily_screen_time_min
social_media_time_min notification_count num_app_switches/VIF
clb;
          title "Appendix B.2";
run;

/*****Part C: Check Validity of Model*****/
```

```
/* Test for Lack of Fit*/
proc reg data= health;
     model digital_wellbeing_score = daily_screen_time_min
social_media_time_min notification_count
num_app_switches/lackfit;
          title "Appendix C.1";
run;

/*Test for Multicollinearity, Outliers, and Influential
Observations*/
proc reg data= health;
     model digital_wellbeing_score = daily_screen_time_min
social_media_time_min notification_count num_app_switches/VIF r
influence;
     output out=diag
          h=leverage
          r = resid
          cookd= cookd
          rstudent= rstudent
          dffits =dffits
          ;
                    title "Appendix C.2";
run;

proc print data = diag;
run;

proc print data = diag;
where dffits > 0.2;
var digital_wellbeing_score daily_screen_time_min
social_media_time_min notification_count num_app_switches
dffits;
run;

proc print data = diag;
where cookd > 4/500;
var digital_wellbeing_score daily_screen_time_min
social_media_time_min notification_count num_app_switches cookd;
          title "Appendix C.3";
run;
*DFBETAS score > 2/sqr(n) = 0.0894427191;
```

```
*DFFITS score > 2sqr(p/n) = 0.2;

Proc reg data = health  plots(only)=(dfbetas dffits);
     model digital_wellbeing_score = daily_screen_time_min
social_media_time_min notification_count num_app_switches;
               title "Appendix C.4";
run;


/* Remdial Measure for Outliers and Influential Points*/
proc robustreg data=diag method=m;
   model digital_wellbeing_score = daily_screen_time_min
                                   social_media_time_min
                                   notification_count
                                   num_app_switches;
     output out=diag_valid;
               title "Appendix C.5";
run;
```